

Symmetry as a Design Axis in Variational Quantum Learning

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Abstract—Symmetry is one of the most powerful priors we can give a learning model: it marks the variations in the input that should leave the answer unchanged, so the model need not learn them from data. The prior is especially valuable in quantum machine learning, where circuits are small and easily overwhelmed, and recent work shows that building a known symmetry into a variational quantum circuit improves generalization. We ask what comes next: once a model is symmetric, is the only choice how much symmetry to impose, or also which symmetry-respecting interactions it may learn? Treating symmetry as a design space, we relax it toward smaller subgroups and find that weaker constraints recover almost all of the benefit, then keep the symmetry but enrich it with trainable interactions acting directly on the patterns that determine the label. This second move lifts performance well beyond the original symmetric circuit, and a control that scrambles the parameter sharing at matched size confirms the gain comes from the task’s geometry, not a smaller model. We demonstrate this on Tic-Tac-Toe board classification, chosen because every relevant symmetry, label, and decisive pattern is known exactly, so the effect can be isolated rather than inferred. A known symmetry is not merely a way to tie parameters; it is a language for deciding which interactions a model may learn, and so it is not the end of architecture design but its beginning.

Index Terms—quantum machine learning, geometric quantum machine learning, equivariance, symmetry, inductive bias, variational quantum circuits, parameter sharing, generalization

I. INTRODUCTION

Symmetry is one of the most useful things we can tell a model in advance. It marks the changes to the input that should not change the answer, so the model does not have to learn this for itself. Turn a picture of a cat sideways and it is still a cat; a model that already knows this need not waste data discovering it. The same idea underlies convolutional networks, which reuse one filter across an image [1], and geometric deep learning, which builds whole architectures around the transformations a task is supposed to ignore [2]. Variational quantum learning models (VQLMs) stand to gain the most from this discipline. They are parameterized quantum circuits trained as statistical models [3], [4], made expressive through data re-uploading [5], [6] yet easily overwhelmed when that expressiveness is unconstrained. Embedding a known symmetry into the circuit resolves the tension: an equivariant ansatz routes input transformations into structured circuit transformations, making invariance a property of the architecture rather than a target of optimization, and improving generalization as a result [7].

We start where that result ends. Once symmetry is in the circuit, the usual remaining question is how much of it to enforce. We argue there is a second, more decisive question that is rarely asked: which symmetry-respecting interactions the circuit is given to learn. Treating symmetry as a design space rather than a switch, we study both axes on a deliberately transparent benchmark, Tic-Tac-Toe board classification, where the relevant symmetry, the labels, and the patterns that decide the outcome are all known exactly and a fixed rule already attains perfect accuracy. The benchmark isolates the mechanism; the game itself is not hard.

We work through three layers. We first confirm that enforcing the full board symmetry improves generalization over an unconstrained circuit. We then relax that symmetry toward smaller subgroups and ask not just how much of the benefit survives, but whether a weaker symmetry can stay on par with the full one. Finally, and most importantly, we keep the symmetry but enrich it, adding trainable gates that act on the three-cell lines that actually decide the game, while still sharing their parameters across the symmetry. The takeaway generalizes well beyond the game: a known symmetry is not just a way to use fewer parameters. It is a language for choosing which interactions a quantum model is allowed to learn.

II. PROTOCOL

Each legal, reachable board is a list of nine values $g_i \in \{-1, 0, +1\}$, one per field, for circle, empty, and cross. Every board falls into one of three classes, circle win, draw, or cross win, written as target vectors $(+1, -1, -1)$, $(-1, +1, -1)$, and $(-1, -1, +1)$. Following the Meyer benchmark, unfinished games count as draws, and train and test splits are class-balanced so accuracy cannot be inflated by a dominant label. We enumerate all 5478 legal boards rather than sampling them, with the full procedure summarized in Appendix A. The label is invariant under the eight symmetries of the square, the four rotations and their reflections, which form the dihedral group D_4 . Figure 1(a) shows this directly: rotating or flipping a board never changes who has won.

To feed a board into the circuit, as in Figure 1(b), we place one qubit on each field and turn each value into a rotation angle $x_i = 2\pi g_i/3$ applied with a single-qubit R_X gate, so circle, empty, and cross land at three evenly spaced angles:

$$U(g) = \bigotimes_{i=0}^8 R_X \left(\frac{2\pi}{3} g_i \right).$$

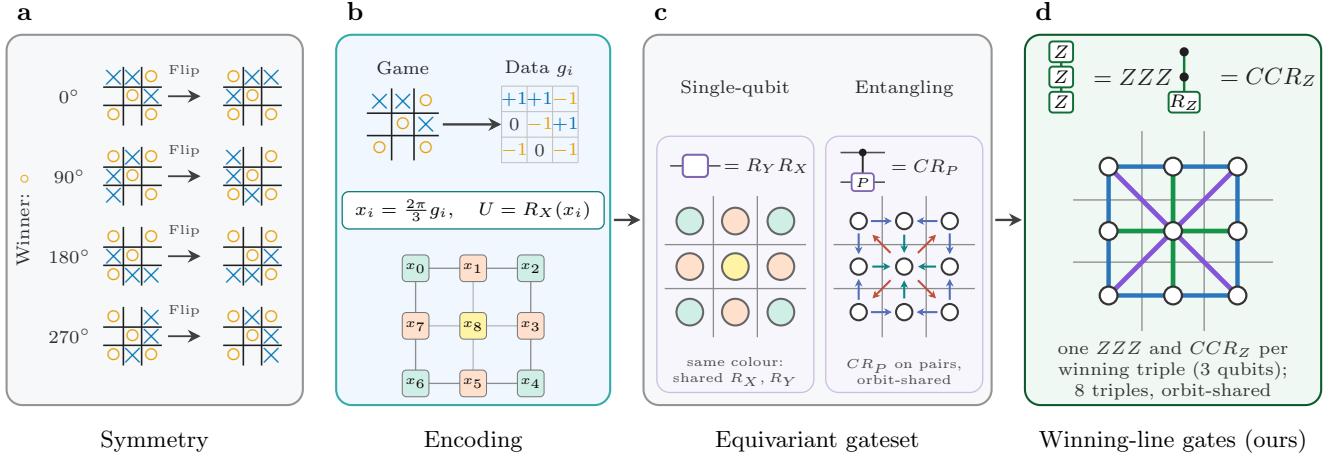


Fig. 1. From task symmetry to a task-aligned equivariant circuit. (a) A Tic-Tac-Toe label is invariant under the eight rotations and reflections of the square, the dihedral group D_4 ; the winner is unchanged throughout. (b) Each field carries a value $g_i \in \{-1, 0, +1\}$ for circle, empty, and cross, mapped to an angle $x_i = \frac{2\pi}{3}g_i$ and encoded on its own qubit via $R_X(x_i)$. Numbering the fields cyclically with the center last makes the D_4 orbits explicit: corners $\{0, 2, 4, 6\}$, sides $\{1, 3, 5, 7\}$, and center $\{8\}$. (c) The original edge ansatz: single-qubit $R_Y R_X$ rotations and controlled CR_P entanglers on neighbouring fields and center connections, with gates in the same orbit sharing one parameter, shown by colour, so the model stays equivariant. (d) Our extension adds orbit-shared gates on the eight winning triples, rows, columns, and diagonals: one ZZZ rotation and one symmetric CCR_Z per triple, enriching the circuit with the motifs that decide the game without breaking the symmetry. Panel layout adapted from Meyer et al. [7].

Numbering the fields cyclically around the rim with the center last, as drawn in the panel, makes the board symmetry explicit. The four corners $\{0, 2, 4, 6\}$, the four sides $\{1, 3, 5, 7\}$, and the center $\{8\}$ are the three D_4 orbits: the symmetries shuffle positions within each set, but never mix one set with another.

The circuit is a stack of identical layers. Each layer re-encodes the board with $U(g)$, a data re-uploading step that lets the circuit express nonlinear functions [5], and then applies a trainable block. Unless stated otherwise, we use $L = 3$ layers with $p = 2$ repetitions of the block per layer. The block is the original edge ansatz shown in Figure 1(c): a single-qubit $R_Y R_X$ rotation on every qubit, followed by controlled CR_P entanglers on neighbouring fields and center connections.

Symmetry enters through how these gates are parameterized. Instead of giving each gate its own angle, all gates within an orbit share one trainable parameter, indicated by colour in Figure 1(c): corners share one single-qubit rotation, sides another, the center keeps its own, and the CR_P entanglers are tied across the orbits of the directed pairs they act on. A rotated or reflected board is then never a new, unrelated input; it permutes the same qubits and reuses the same parameters, which is precisely what makes the model equivariant. The readout follows the same geometry: we average the Pauli- Z expectation over each orbit to obtain a three-number vector for corners, center, and sides, and take its largest component as the predicted class.

Because the orbits are defined by whichever symmetries we choose to enforce, we can dial the symmetry up or down without touching the wiring. For a subgroup $H \leq D_4$,

parameters are tied across the orbits induced by H , leaving the circuit topology fixed and changing only the sharing pattern. Comparing no sharing, the two Z_2 subgroups, C_4 , D_2 , and full D_4 gives 204, 108, 126, 60, 78, and 54 trainable parameters for the edge ansatz at $L = 3, p = 2$, respectively; in this family, more symmetry means fewer free parameters.

The edge ansatz only couples neighbouring fields, yet a win is decided by a full three-cell line. This gap motivates the extension in Figure 1(d). We add trainable gates directly on the eight winning triples, the three rows, three columns, and two diagonals shown in blue, green, and purple, using one ZZZ rotation and one symmetric CCR_Z interaction per triple. We test each channel on its own and together. Our main model, edge+lines, keeps the original edge ansatz and adds both line-gate channels. Crucially, these new gates are shared over the same orbits, so the model remains equivariant under the chosen subgroup. At $L = 3, p = 2$, the full- D_4 edge model has 54 parameters and the full- D_4 edge+lines model has 90, still far below the 204 of the unshared edge baseline.

All reported runs use mean-squared error, Adam with learning rate 0.01 [8], batch size 15, 30 minibatch updates per epoch, 100 epochs, fixed seed blocks, PennyLane simulation [9], and PyTorch optimization [10]. Exploratory runs outside this protocol are not used for any headline claim.

III. RESULTS

Figure 2 shows the main empirical story. In panel (a), the unconstrained edge ansatz carries many more parameters yet generalizes worse than the D_4 -equivariant edge model. At 600

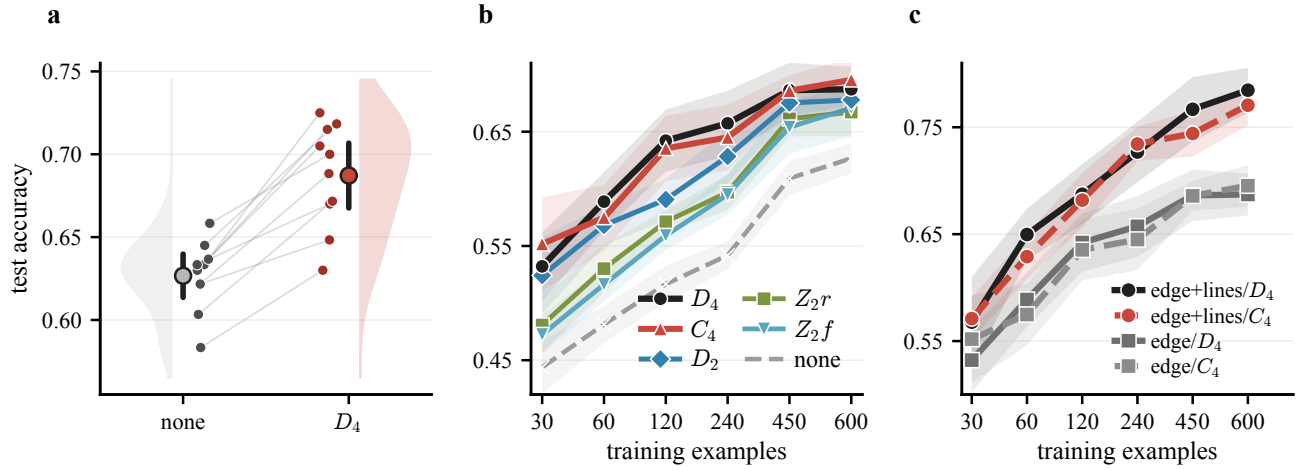


Fig. 2. Main results. (a) Full D_4 improves over no sharing. (b) C_4 closely follows D_4 across train sizes. (c) Edge+lines gives the largest gain while preserving equivariance. Error bars are 95% confidence intervals over seeds.

training examples, edge/none reaches about 0.63 test accuracy while edge/ D_4 reaches about 0.69 with a smaller train-test gap. This reproduces the central lesson of Meyer et al.: the correct board symmetry removes irrelevant functions from the hypothesis class rather than simply shrinking the model.

Panel (b) makes the picture less binary. Full D_4 is the conservative default, but C_4 tracks it closely across the data grid: it is slightly ahead at 30 examples, and from 120 through 600 examples the two stay essentially together while weaker or unconstrained sharing lags. We read this cautiously. It does not establish that partial symmetry is generally better; it shows that symmetry strength is a tunable bias, with rotation-only equivariance preserving most of the benefit while relaxing reflection constraints. The random-sharing control in fig. 3(a) is the guardrail: it fixes parameter count but scrambles the group-orbit structure, and performs substantially worse than orbit-based sharing at every subgroup. The useful prior is the board geometry, not just the smaller circuit.

The strongest effect, in fig. 2(c), comes from enriching the symmetry rather than weakening it. The original edge/ D_4 model sits near 0.69; adding both winning-line channels to the same equivariant circuit lifts the best D_4 model to roughly 0.80, with the C_4 variant close behind. This is the central result: the gain comes not from abandoning symmetry, but from giving the equivariant ansatz direct access to the motifs that decide the label.

Figure 3 rules out two simpler explanations. The gain is not arbitrary parameter reduction, since matched-size random sharing falls short of orbit sharing in panel (a). It is not simply more parameters either: edge+lines/ D_4 has 90 parameters, more than edge/ D_4 at 54 but far below the 204 of the unshared baseline. Panel (c) makes the point geometric: accuracy does not follow parameter count, and edge+lines models occupy a higher band at modest size. The ablation in panel (b) confirms the composition: line-only circuits help, but the strongest model combines local edge structure with both winning-line channels.

IV. DISCUSSION

The results support an architectural view of equivariance. Meyer et al. showed that full task symmetry improves a VQLM; we keep that lesson but show that equivariance is only the starting point. Once a model is symmetric, the next choice is which symmetry-compatible interactions it can learn.

The subgroup results do not prove that C_4 is generally better than D_4 . They show that partial equivariance is a meaningful modeling axis: weaker symmetry can retain most of the benefit while giving the circuit slightly more freedom. The safe conclusion is competitiveness, not dominance.

The stronger conclusion is edge+lines. A pairwise edge ansatz must assemble winning triples indirectly; line gates expose those triples directly while sharing parameters over the same board symmetries. The point is not fewer weights alone, but weights placed on the right motifs.

Together, this supports treating symmetry as a design axis in variational quantum learning. Relaxing full D_4 symmetry to C_4 diagnoses overconstraint by testing whether imposed equivariance is too restrictive for the data and circuit family. Here, however, the larger gain comes from enriching the equivariant ansatz with task-aligned interactions. When such a circuit underperforms, ask not only whether the symmetry is too strong, but whether the circuit exposes the motifs that determine the label.

Soft symmetry is a natural next step: a controlled symmetry-breaking residual could interpolate between hard equivariance and a free ansatz. Such relaxations should be compared against enriched equivariant gates, not only against the original edge circuit.

Tic-Tac-Toe is small, enumerable, and solved perfectly by a deterministic rule oracle, so its value here is as a controlled setting in which the effect of architectural choices can be isolated. The narrower claim is that task-aligned equivariant gates can improve generalization while preserving symmetry. For larger benchmarks, the transferable recipe is to identify

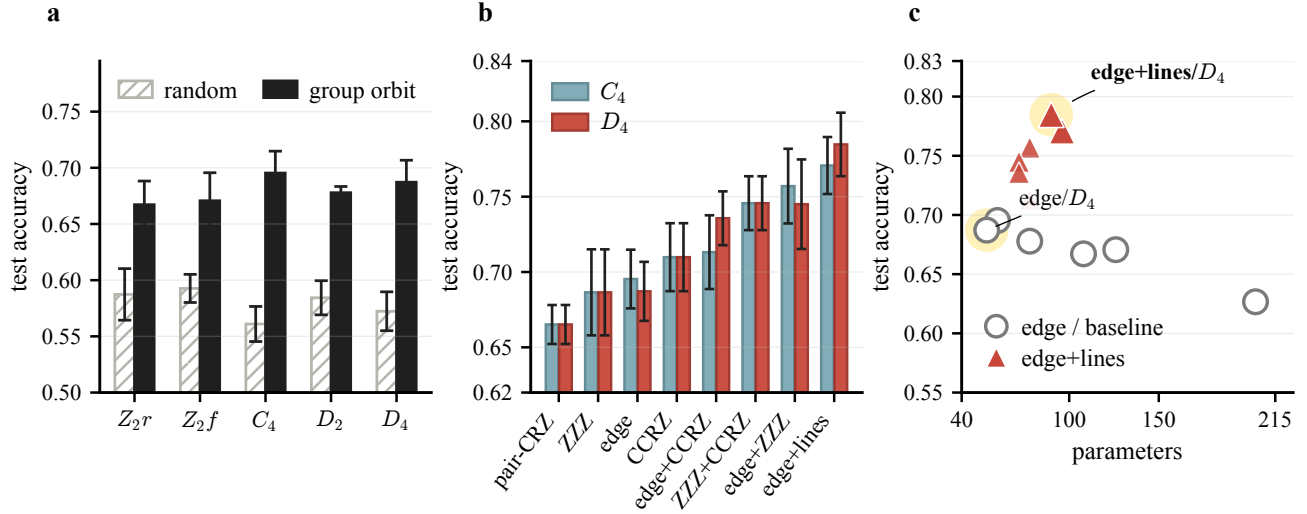


Fig. 3. Controls and ablations. (a) Parameter-matched random sharing underperforms group-orbit sharing. (b) Ablations show that the strongest model combines edge interactions with both line channels. (c) Accuracy is not explained by parameter count alone.

invariant motifs and encode them as orbit-shared trainable interactions.

The original question was whether to impose symmetry. Our results make it richer: symmetry helps, its strength is a modeling axis, and symmetry-compatible interactions can matter most. A known symmetry is a vocabulary: it tells us which parameters to tie and which interactions are worth building. Equivariant-circuit design begins, not ends, once the symmetry is fixed.

APPENDIX A: DATA GENERATION

Data generation and sampling. We follow the Tic-Tac-Toe benchmark protocol of Meyer et al. Legal reachable board states are generated by expanding the game tree from the empty grid, alternating cross and circle moves, and stopping once a game is decided. This yields 5478 distinct boards, stored as cyclically indexed values $g_i \in \{-1, 0, +1\}$ with the center last so the corner, side, and center fields match the D_4 orbits. Labels follow the winning-line rule, with unfinished positions counted as draws. For every seed and training size we draw class-balanced train and test subsets and reuse the same split for all models compared at that setting. Meyer et al.’s headline Tic-Tac-Toe setup uses 450 training examples and a fixed test set of 600 examples; our sweep keeps the same balanced-sampling rule and test size while varying the number of training examples.

Reproducibility. Code, figure sources, and checked CSVs are available at github.com/eybmits/qc_symmetry. All reported VQLM runs use one protocol: $L = 3$, $p = 2$, the paper single-qubit block, mean-squared error, Adam at learning rate 0.01, batch size 15, test size 600, train sizes 30, 60, 120, 240, 450, 600, seeds 0–9, and 100×30 optimizer updates. Figures regenerate with `python paper/make_figures.py`; the manuscript rebuilds with

`latexmk -pdf -interaction=nonstopmode main.tex`. Validation requires 360 edge rows, 120 edge+lines rows, 160 ablation rows, and 110 random-sharing rows; intervals are 95% confidence intervals, and the rule oracle is a label sanity check, not a plotted baseline.

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